Practical 2.3 & 2.4: CRD Customisation and Data Layers

Branding, settings, Mwanza/Tanga data upload, QGIS WMS/WFS connections

Day 2 · 3 hours · Resilience Academy SDI Training

Reference: `crd.resilienceacademy.ac.tz` | Data: Tanga & Mwanza, Tanzania

Learning Objectives:

- Customise your CRD instance with your institution's branding and settings
- Upload Mwanza and Tanga datasets from the live CRD
- Set metadata and permissions on uploaded layers
- Validate layers in QGIS via WMS/WFS connections

CRD Customisation — Django Admin

```
# Access Django admin at http://localhost/en/admin
# Login with your superuser credentials

# In admin panel:
# Sites > Sites > example.com
# Change to: crd-training.local
# GeoNode > Theme Settings
# SITE_NAME = "Tanzania SDI Training CRD"
```

CRD Customisation — local_settings.py

```
# Edit my_geonode/local_settings.py

SITE_NAME = "Tanzania SDI Training CRD"
SITE_HOST_NAME = "localhost"

THEME_ACCOUNT_CONTACT_EMAIL = "admin@resilienceacademy.ac.tz"

# Default map centred on Mwanza
DEFAULT_MAP_CENTER = (32.9, -2.5)
DEFAULT_MAP_ZOOM = 12

# To centre on Tanga instead:
# DEFAULT_MAP_CENTER = (39.1, -5.07)
# DEFAULT_MAP_ZOOM = 12

# Allow public to view layers
DEFAULT_ANONYMOUS_VIEW_PERMISSION = True
DEFAULT_ANONYMOUS_DOWNLOAD_PERMISSION = False
```

```
# Restart Django to apply settings
docker compose restart django
```

Download Mwanza Data from Live CRD

```
mkdir ~/crd-data && cd ~/crd-data

# Download Mwanza businesses (WFS GeoJSON)
curl -s "https://crd.resilienceacademy.ac.tz/geoserver/ows?\
service=WFS&version=2.0.0&request=GetFeature\
&typeName=geonode:mwanza_businesses\
&outputFormat=application/json" \
  -o mwanza_businesses.geojson

# Download Mwanza education facilities
curl -s "https://crd.resilienceacademy.ac.tz/geoserver/ows?\
service=WFS&version=2.0.0&request=GetFeature\
&typeName=geonode:mwanza_education_facilities\
&outputFormat=application/json" \
  -o mwanza_education_facilities.geojson

# Check downloads
python3 -c "
import json
for f in ['mwanza_businesses.geojson','mwanza_education_facilities.geojson']:
    with open(f) as fp:
        d = json.load(fp)
    print(f'{f}: {len(d[\"features\"])} features')
"
```

Upload Layers to Your Local CRD

1. Open `http://localhost` and log in as admin
2. Click **Data** → **Upload Layer**
3. Select `mwanza_businesses.geojson` and click Upload
4. Fill in metadata:
    - Title: `Mwanza Businesses`
    - Abstract: `Business locations in Mwanza city, Tanzania. Exposure data for urban resilience assessment.`
    - Keywords: `mwanza, businesses, exposure, tanzania`
    - Category: **Exposure**
5. Repeat for `mwanza_education_facilities.geojson`

Load Tanga Imagery in QGIS via WMS

```
# In QGIS:
# Layer > Add Layer > Add WMS/WMTS Layer
# New connection:
# Name: Resilience Academy CRD
# URL: https://crd.resilienceacademy.ac.tz/geoserver/ows?service=WMS
# Connect > Search for: Tanga_imagery_reduced
# Add layer to map

# Also add Mwanza layers:
# Layer > Add WFS Layer
# URL: https://crd.resilienceacademy.ac.tz/geoserver/ows?service=WFS
```

```
# Connect > find mwanza_businesses > Add
```

Set Layer Permissions

1. Go to the Mwanza Businesses layer page
2. Click **Edit** → **Layer Info**
3. Under **Permissions**, set:
   - Anyone can view: **checked**
   - Registered users can download: **checked**
   - Only owner can edit: **checked**

Checkpoint:

✓ Your CRD site name changed successfully
✓ Default map centred on Mwanza
✓ `mwanza_businesses` layer visible on your local CRD map
✓ `mwanza_education_facilities` layer uploaded with metadata
✓ QGIS shows Tanga satellite imagery from live CRD WMS
✓ Mwanza business points visible in QGIS via WFS